Symbolic and Distributed AI
An Introduction to Agents and MASs

Viviana Mascardi

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Dibris

Outline

1. What are Agents and MASs
   - What are agents and MAS
   - MASs for software engineering
   - MASs for development of software applications
   - MASs for simulation
2. Short history of agents and MASs

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An agent is an hardware or software system

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- flexible
  - reactive
  - proactive
  - social

What is an agent?







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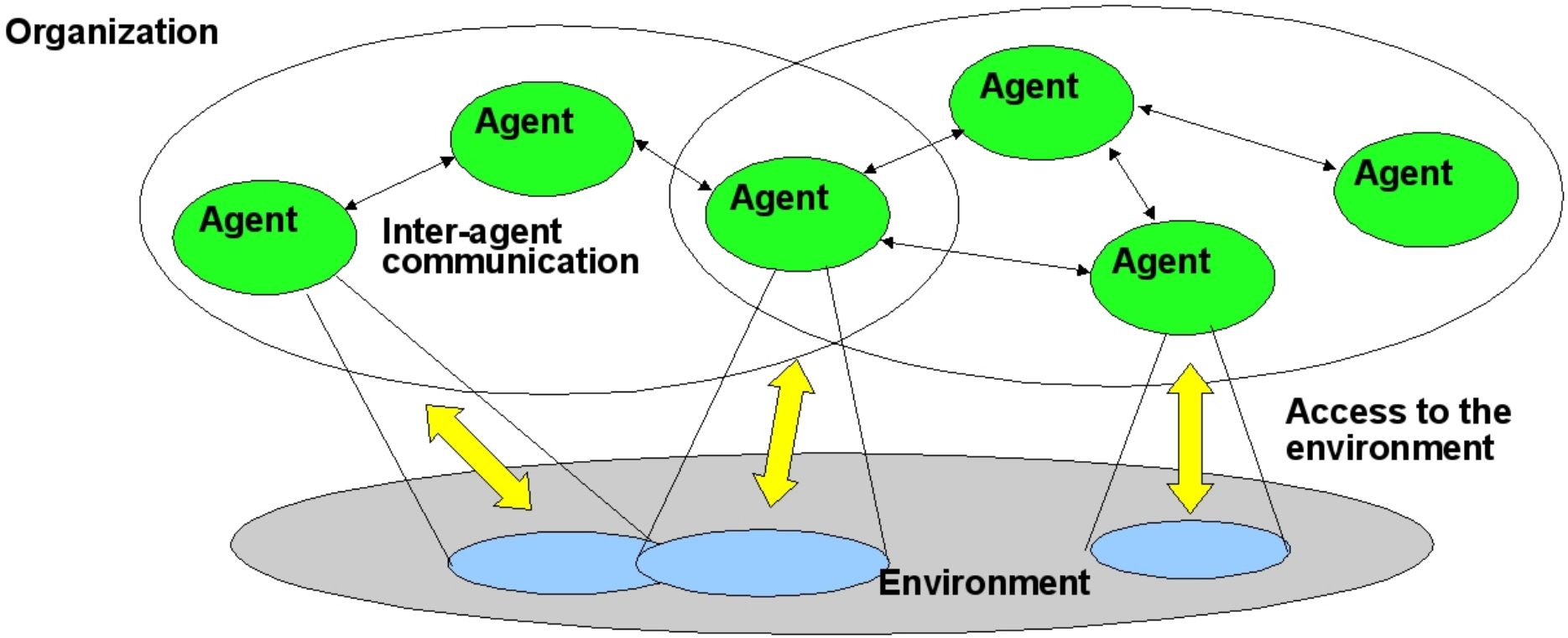
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  Mentalistic notions: beliefs, desires, intentions, commitments, . . .
- *J. Bates, Communications of the ACM, 37(7), 1994*
  Emotional notions: friendlyness, trust, untrust; agents must be believable

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What is a MAS?


Organization
Agent
Agent
Agent
Agent
Agent
Agent
Inter-agent
communication
Access to the
environment
Environment

What is a MAS?



Franco Zambonelli



Andrea Omicini

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MASs as a Software Engineering metaphor for analysis and design

F. Zambonelli, A. Omicini, JAAMAS, 9(3), 2004

"...today's software engineering approaches are increasingly adopting abstractions approaching that of agent-based computing.

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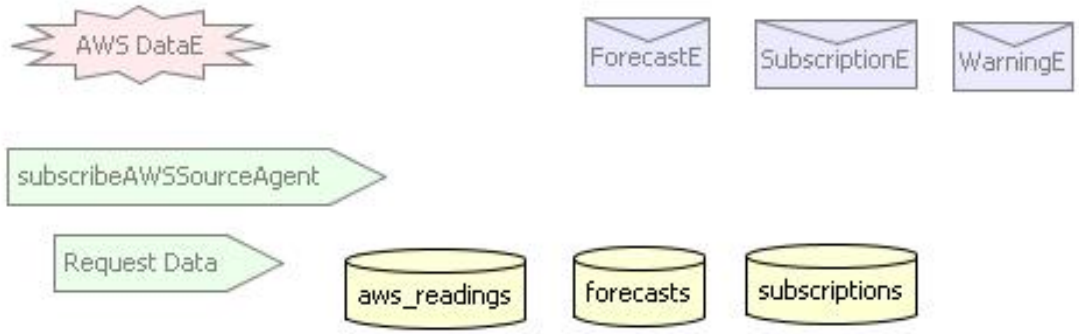
MASs as a Software Engineering metaphor for analysis and design

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"...today's software engineering approaches are increasingly adopting abstractions approaching that of agent-based computing.
This trend can be better understood by recognising that the vast majority of modern distributed systems scenarios are intrinsically prone to be developed in terms of MASs, and that modern distributed systems are already de facto MASs, i.e., they are indeed composed of autonomous, situated, and social components"

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MASs as a Software Engineering metaphor: how?



PDT, `http://www.cs.rmit.edu.au/agents/pdt/`

MASs as a Software Engineering metaphor: how?

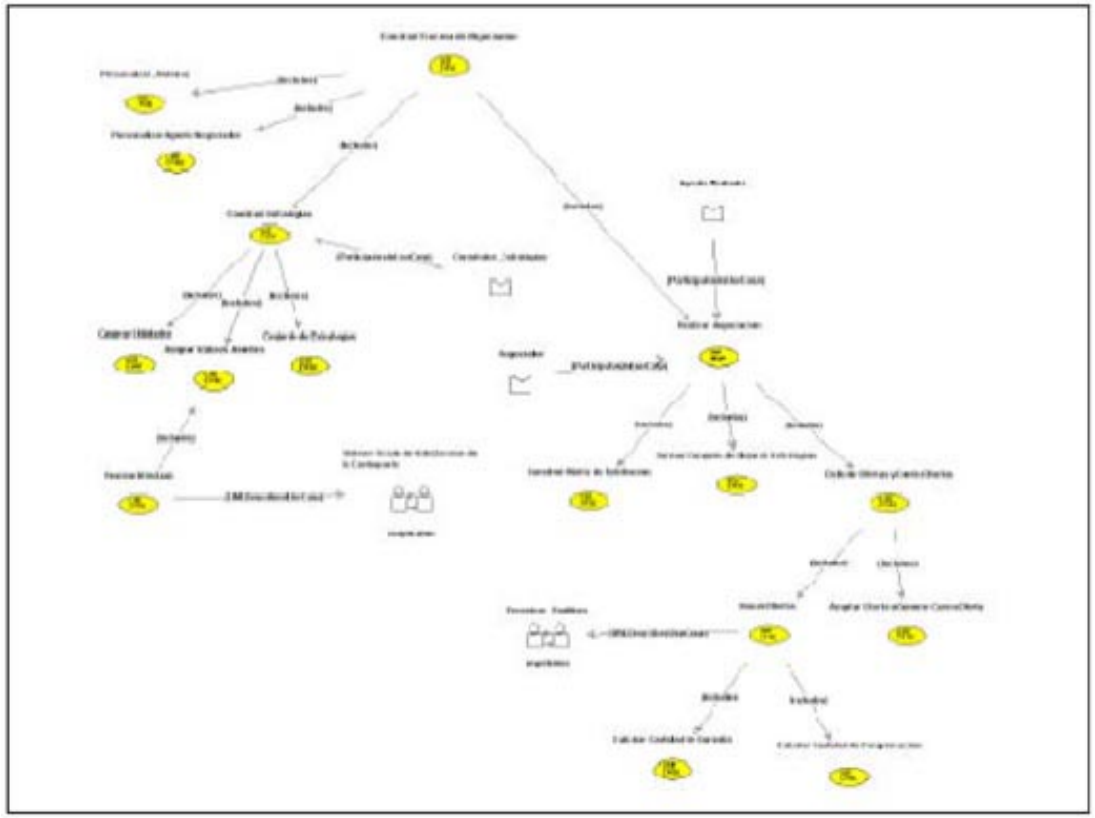


Figura 4. Funcionamiento General del Sistema de Negociación Automatizada

Ingenias, `http://ingenias.sourceforge.net/`

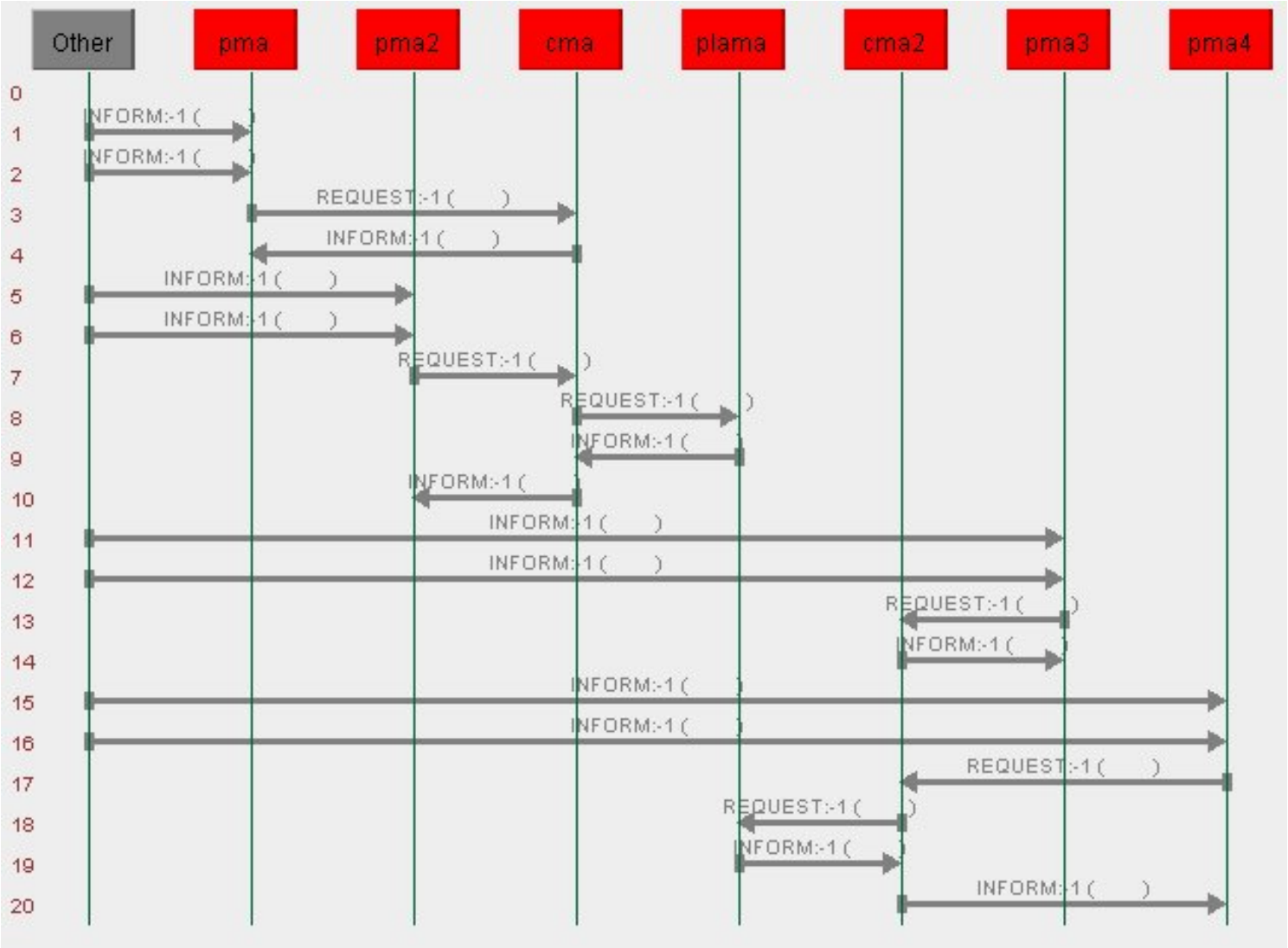
Agent-based development

- Agents are employed as the technology for *implementing the software application itself*.

Agent-based development

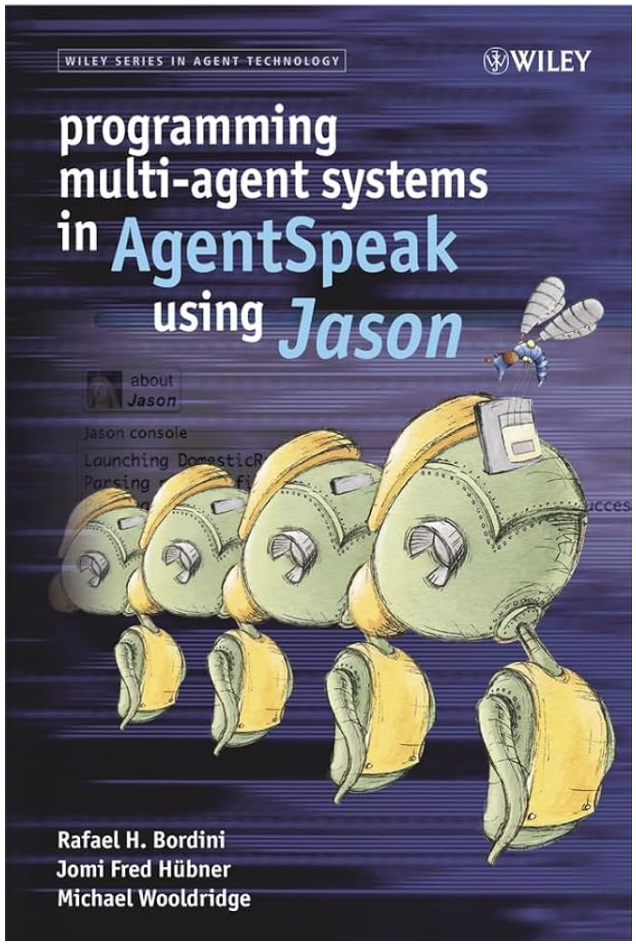
- Agents are employed as the technology for *implementing the software application itself*.
- This choice is usually made when the agent metaphor has already been applied for the engineering steps coming before in the SW life cycle, when the intelligence of the individual agent is more relevant than the “emerging intelligence” of the overall system, when complex patterns of interactions may be required, when the application to be developed is geographically distributed, is decentralized, is open.

Agent-based development: how?



Jade, `http://jade.tilab.com/`

Agent-based development: how?



Jade, `http://sourceforge.net/projects/jason/`

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Agent-based development

Many agent-based development tools exist, but they are not commercial projects supported by large and solid companies. This is one of the reasons for the limited adoption of this technology by industries.

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Many agent-based development tools exist, but they are not commercial projects supported by large and solid companies. This is one of the reasons for the limited adoption of this technology by industries.
But, what about using existing technologies that are not officially tagged as “agent-oriented”, but that are very suitable for implementing software agents, for the implementation stage, and a “pure” agent approach for the modeling and design stages?

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Agent-based development: agents and SOC (2009)

University of South Carolina
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1-1-2009

Agents and Service-Oriented Computing for Autonomic Computing: A Research Agenda

Frances M.T. Brazier

Jeffrey O. Kephart

H. Van Dyke Parunak

Michael N. Huhns
University of South Carolina - Columbia, huhns@sc.edu

Agent-based development: agents and cloud (2011)

Software Agents as Cloud Computing Services

FULL TEXT

CONFERENCE PAPER · JANUARY 2011 *with* 50 READS

DOI: 10.1007/978-3-642-19875-5_35 · Source: DBLP

Conference: Advances on Practical Applications of Agents and Multiagent Systems - 9th International Conference on Practical Applications of Agents and Multiagent Systems, PAAMS 2011, Salamanca, Spain, 6-8 April 2011

1st Ignacio Lopez-Rodriguez
8.58 · Instituto Tecnológico de Canarias

2nd Mario Hernández-Tejera

ABSTRACT

The community devoted to the agency theory needs to create practical solutions capable of representing users in the virtual societies which are emerging as a result of Internet. Cloud Computing has precisely succeeded as a model able to bring software solutions to users in a practical and transparent manner. To foster the adoption of agent-based solutions by the users, this paper describes a model where software agents figure as a new Cloud Computing service which would represent clients in virtual environments. We discuss the challenges that entail the proposal, the technologies necessary for its implementation and finally we develop a proof of concept to confirm its viability.

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Agent-based development: agents and cloud (2011)

Cloud Computing and Software Agents: Towards Cloud Intelligent Services

Domenico Talia
ICAR-CNR & University of Calabria
Rende, Italy
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Abstract — **Cloud computing systems provide large-scale infrastructures for high-performance computing that are "elastic" since they are able to adapt to user and application needs. Clouds are used through a service-oriented interface that implements the *-as-a-service paradigm to offer Cloud services on demand. This paper discusses Cloud computing models and architectures, their use in parallel and distributed applications, and examines analogies, differences and potential synergies between Cloud computing and multi-agent systems. This analysis is lead having in mind the goal of implementing high-performance complex systems and intelligent applications by using of Cloud systems and software agents. The convergence of interests between multi-agent systems that need reliable distributed infrastructures and Cloud computing systems that** run on a parallel or distributed computer to achieve the needed high performance for solving large complex problems keeping execution time low.

Although several differences exist between Cloud computing and multi-agent systems, they are two distributed computing models, therefore several common problems can be identified and several benefits can be obtained by the integrated use of Cloud computing systems and multi-agents. The research activities in the area of Cloud computing are mainly focused on the efficient use of the computing infrastructure, service delivery, data storage, scalable virtualization techniques, and energy efficiency. In summary, we can say that in Cloud computing the main focus of research is on the

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Agent-based development: agents and WSs (2013)

JADE WEB SERVICES INTEGRATION GATEWAY (WSIG) GUIDE

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Version: 3.0

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Authors: JADE Board

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Agent-based development: agents and IoT (2013)

From Internet of Things to Internet of Agents

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***Abstract*—From sophisticated single agent in complex environments to multi-agent system (MAS) organizations, intelligent software agent research has come a long way in just under two decades. Many new branches of research in this field have emerged over the years which have enabled today's agents to perform a wide variety of human-like tasks such as learning, reasoning, negotiating, self-organizing and trusting each other, etc. Unfortunately, very few practical MASs have been deployed after such a long period of intensive research and development. For MASs to achieve higher popularity among end-users, we believe that agent oriented software engineering (AOSE) should adopt a new paradigm as has been done in Web 2.0 - to allow end-users to actively participate in developing or modifying features in agents at various stages of the agent's lifecycle. In this paper, we propose a vision for democratizing AOSE. We discuss what potential new researches need to be carried out in the areas of AOSE and agent learning in order to realize such a vision of moving from MASs to mass end-user agent development, and discuss potential challenges facing various aspects of this vision.**

I. Introduction

As Internet of Things (IoT) become strives towards the fusion of the physical, social and cyber spaces, the nodes are increasingly required to exhibit intelligent behaviors to varying degrees depending on the domain of application. It is natural to view and implement them as software agents in an IoT through Agent Oriented Software Engineering (AOSE). The AOSE paradigm has come a long way since its inception. Over

Track in Innovative Applications in the International Conference on Autonomous Agents and Multi-Agent Systems (AAMAS), most of the agent applications published so far are targeted for usages that not often actively involve the general public as direct stakeholders who actively interact with the technology [4], [5]. This is somehow not entirely the case for applications in the domain of education where personal interactions between the agents and learners frequently occur [6], [7]. However, in these cases, the interactions are still limited by predefined processes which have been incorporated into these agents.

Compared to the field of MAS, Web 2.0 [8] has rapidly transformed the Internet within a short span of less than 10 years. Today, millions of users interact and create content and applications for Web 2.0 based platforms such as Wikipedia, Youtube, Facebook, Second Life and Twitter etc. Together with the paradigm of end-user development [9], [10], these Web 2.0 platforms have grown enormously successful by involving end-users as an auxiliary developer force into their digital ecosystems. It was estimated that in 2005, there were 55 million end-user developers in the U.S. alone as compared to 2.75 million professional developers. It is an active environment which can be viewed as crowdsourcing on steroid. By opening up the development of contents and applications for end-users, the Web 2.0 concept effectively democratized the Internet and created an ideal environment for explosive

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Agent-based development: agents and IoT (2015)

Agents of Things (AoT):

An Intelligent Operational Concept of the Internet of Things (IoT)

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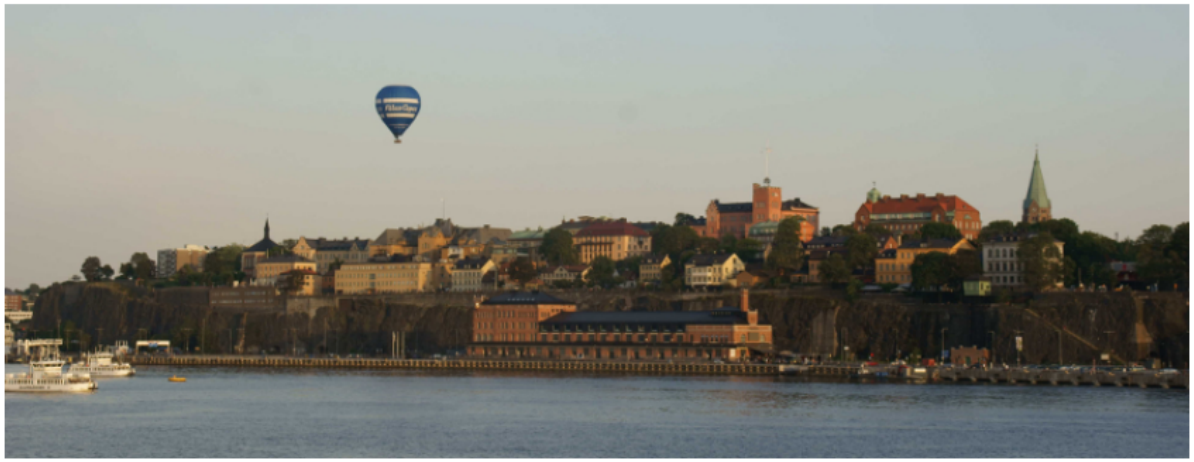
***Abstract*—In this conceptual paper, we review the definitions, characteristics, and architecture of the Internet of Things (IoT) concept. We then identify the deficiencies of the IoT concept, analyze them, and discover the issue of lack of reasoning and intelligence in the IoT concept. We propose a solution to augment the IoT with intelligent software agents resulting in a new concept called the Agents of Things (AoT). The paper presents the AoT architecture and a scenario for one application of the proposed concept. Finally, it discusses the benefits of implementing AoT concept to solve real world**

next phase of Internet evolution to achieve the goal of machine to machine (M2M) learning [7]. Other researchers consider the IoT as an expansion of Internet-based network [3]. However, there is a major difference between IoT and the Internet. The next few sections highlight the IoT and its difference from the Internet.

A. The Concept of IoT

The IoT is different from the Internet in communication aspect. The Internet communication can be established at any

Agent-based development: agents and IoT (2018)



Papers accepted for presentation & keynote

KEYNOTE: Simon Mayer, TU Graz and Pro²Future GmbH

Title: Autonomous Agents for Flexible Hypermedia Systems (EMAS18-invited-talk-slides.pdf)

Abstract: The Web of Things community used to be driven by the application of Web technologies to enable flexible mashups of smart devices on top of the Internet of Things, an objective that we consider accomplished (from a research standpoint) in many different domains ranging from smart homes and cars to dynamic factories in the Industrie 4.0 paradigm. One of the next big things for us – consequently, perhaps, from an AAMAS standpoint – is to increase the autonomy of our Web-enabled devices and their understanding of one another, for instance by outfitting them with semantic descriptions of their properties and functions and, sometimes, even bestowing agency upon them. In this talk, I discuss this convergence that will enrich real-world devices with AAMAS technologies, and open up real-world applications to the AAMAS community, while examining important properties of the Web architecture that support flexibly interacting autonomous things on the Web.

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Agent-based development: agents and VR (2023)

Open Access Article

VEsNA, a Framework for Virtual Environments via Natural Language Agents and Its Application to Factory Automation †

by Andrea Gatti and Viviana Mascardi *

Department of Informatics, Bioengineering, Robotics and Systems Engineering (DIBRIS), University of Genoa, Via Dodecaneso 35, 16146 Genoa, Italy

* Author to whom correspondence should be addressed.

† This Paper Extends the Work in Gatti, A.

Robotics **2023**, *12*(2), 46; **https://doi.org/10.3390/robotics12020046**

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(This article belongs to the Special Issue **Agents and Robots for Reliable Engineered Autonomy 2023**)

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Agent-based development: agents and XR (2024)



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Agent-based simulation

C. M. Macal and M. J. North, Proc. of Winter Simulation Conference, 2007

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"... Agent-based modeling and simulation (ABMS) is a new approach to modeling systems comprised of autonomous, interacting agents. ABMS promises to have far-reaching effects on the way that businesses use computers to support decision-making and researchers use electronic laboratories to support their research."

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"Some have gone so far as to contend that ABMS "is a third way of doing science," in addition to traditional deductive and inductive reasoning.

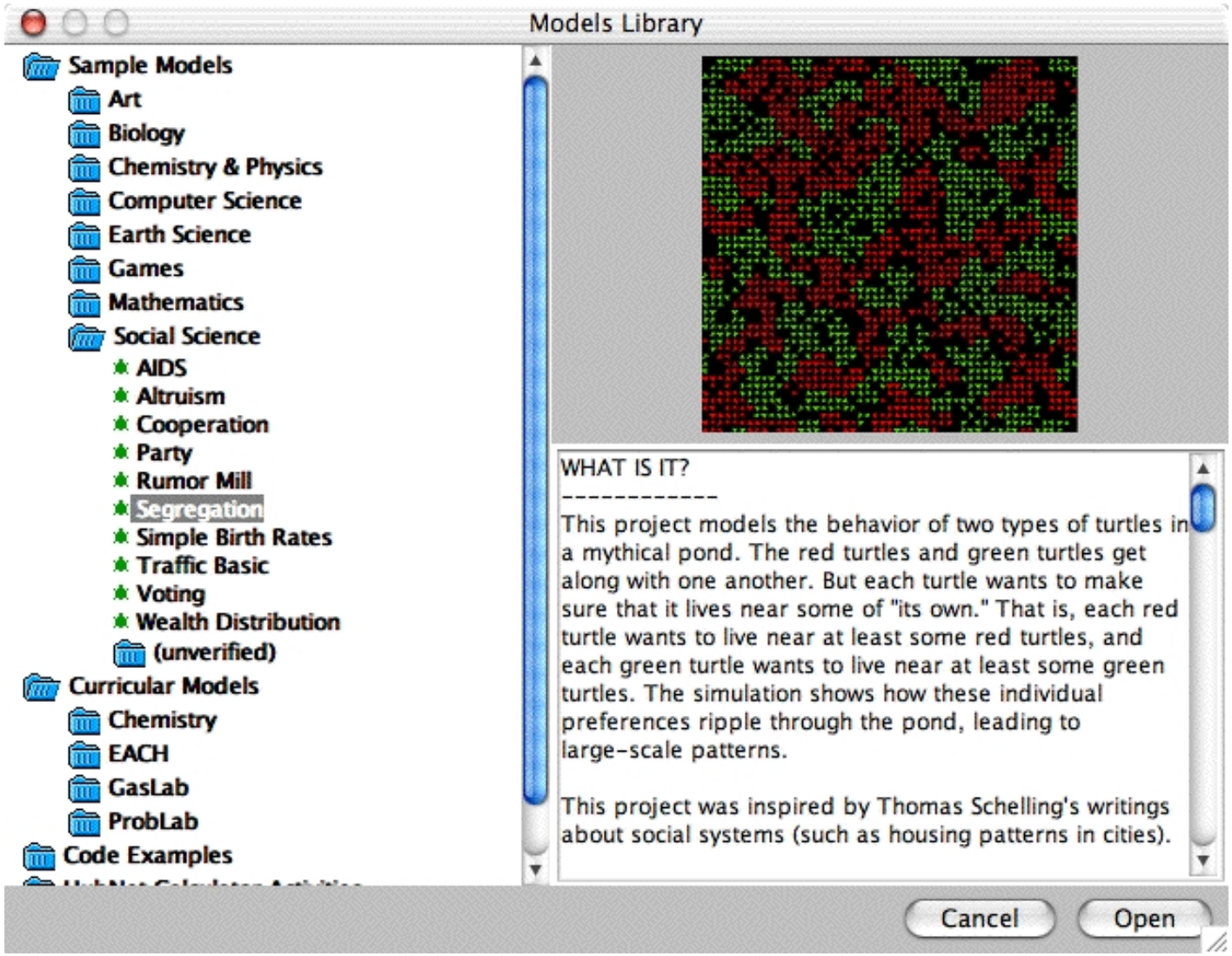
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"Some have gone so far as to contend that ABMS "is a third way of doing science," in addition to traditional deductive and inductive reasoning.
Computational advances have made possible a growing number of agent-based models across a variety of application domains.
Applications range from modeling agent behavior in the stock market, supply chains, and consumer markets, to predicting the spread of epidemics, the threat of bio-warfare, and the factors responsible for the fall of ancient civilizations. "

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Agent-based simulation: how?



NetLogo, `http://ccl.northwestern.edu/netlogo/`

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Agents' history

Current interest in autonomous agents did not emerge from a vacuum. Researchers and developers from many different disciplines have been talking about closely related issues for some time. The main contributors are:

- artificial intelligence
- object-oriented programming and concurrent object-based systems
- human-computer interface design

Artificial Intelligence

- Among the oldest areas of research, the activity most closely connected with that of autonomous agents was AI “planning” (*A. Newell e H. A. Simon, Lernende Automaten, 1961; R. E. Fikes e N. Nilsson, Artificial Intelligence, 5(2), 1971*).
- Until the 1980s comparatively little effort within the AI community was directed to the study of intelligent agents. The primary reason for this state of affairs was that AI researchers had historically tended to focus on the various different components of intelligent behaviour (learning, reasoning, problem solving, vision understanding and so on) in isolation.
- Expert systems: “disembodied intelligence”, no real-time capabilities, no social ability.

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Artificial Intelligence

- First delusions with planning: “first-principle planning” (meaning that, in order to satisfy a goal, the agent has to formulate an entirely new plan for that goal) does not allow the implementation of reactive agents (“calculative rationality”) and it is undecidable in many cases (*D. Chapman, Artificial Intelligence, 32, 1987*)
- Traditional, symbolic approaches to agency and AI leave room to the “behavioural AI”, or “reactive AI” (*R. Brooks, Artificial Intelligence, 47, 1991*).

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Artificial Intelligence

- From the ’90s, researchers start to recognise that both purely reactive architectures, à la “Brooks”, and purely deliberative architectures, à la “first-principle planning”, have both advantages and disvantages ==> birth of “hybrid architectures”, usually stratified (*TouringMachines by I. A. Ferguson, 1992, InteRRaP by J. P. Muller e M. Pischel, 1994*).
- Practical reasoning agents: agents whose architecture is modelled on or inspired by a theory of practical reasoning in humans. By practical reasoning, we mean the kind of pragmatic reasoning that we use to decide what to do.
  The most widely known architecture for practical reasoning is the Belief-Desire-Intention one (*M. P. Georgeff e A. L. Lansky, Proc. di AAAI-87, 1987*).

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Artificial Intelligence

- 1993: researchers start to understand and accept that intelligent agents are not just a branch of AI, but they are an autonomous technological and methodological approach. Y. Shoham, with his paper *"Agent-oriented programming", Artificial Intelligence 60(1)*, is considered the father of agent oriented programming.

Object-oriented programming

- Objects are defined as computational entities that encapsulate some state, are able to perform actions, or methods on this state, and communicate by message passing.
- So, what is the difference between an agent and an object?

Object-oriented programming

Differences between agents and objects:

- *Autonomy*: an object can be thought of as exhibiting autonomy over its state: it has control over it. But an object does not exhibit control over it's behaviour. "*Objects do it for free; agents do it for money*".
- *Flexibility*: the standard object model has nothing whatsoever to say about how to build systems that integrate proactive, reactive and social behaviour. One could argue that we can build object-oriented programs that *do* integrate these types of behaviour. But this argument misses the point, which is that the standard object-oriented programming model has nothing to do with these types of behaviour.

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Human-computer interface design

With respect to the interaction paradigm known as direct manipulation, we would like to have programs that behave like “digital butlers”:

N. Negroponte, Being Digital, 1995

- *“The agent answers the phone, recognizes the callers, disturbs you when appropriate, and may even tell a white lie on your behalf”.*
- *“If you have somebody who knows you well and shares much of your information, that person can act on your behalf very effectively. If your secretary falls ill, it would make no difference if the temping agency could send you Albert Einstein. This issue is not about IQ. It is shared knowledge and the practice of using it in your best interests”.*
- *“Like an army commander sending a scout ahead ... you will dispatch agents to collect information on your behalf”.*

Intentional Systems

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- Janine took her umbrella because she *believed* it was going to rain.
- Michael worked hard because he *wanted* to possess a PhD.

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[D. C. Dennett, The Intentional Stance, 1989]

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Agents as Intentional Systems

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- Modal logic languages are suitable for **specifying** agents as intentional systems.
- Languages based on computational (modal) logic are suitable for **programming agents**.
- Axiomatised logic-based languages can undergo an axiomatic **verification**; other languages which can be used for **model checking**.

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Modal Logic

Modal logic is an extension of classical logic with (generally) a new connective $\Box$ and its derivable counterpart $\Diamond$, known as *necessity* and *possibility* respectively.

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If a formula $\Box p$ is true, it means that p is necessarily true, i.e. true in every possible scenario, and $\Diamond p$ means that p is possibly true, i.e. true in at least one possible scenario.

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It is possible to define $\Diamond$ in terms of $\Box$:

$$\Diamond p \Leftrightarrow \neg \Box \neg p$$

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Modal Logic

Different kinds of modal logics exist:

- epistemic logic
- temporal logic
- deontic logic
- dynamic logic
- ... and combinations of them (BDI logic, KARO logic, ...)

History of MASs: the Contract Net Protocol

In the Contract Net Protocol (*Davis, R. and R. G. Smith. Negotiation as a metaphor for distributed problem solving. Artificial Intelligence, vol. 20 pp. 63-100, 1983*), agents can dynamically take two roles: manager or contractor.

- Given a task to perform, an agent first determines whether it can break it into subtasks that could be performed concurrently.
- It employs the Contract Net Protocol to announce the tasks that could be transferred, and requests bids from nodes that could perform any of these tasks.
- A node that receives a task announcement replies with a bid for that task, indicating how well it thinks it can perform the task.
- The contractor collects the bids and awards the task to the best bidder.

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History of MASs: Actors

Actors (*C. Hewitt, P. Bishop and R. Steiger. A Universal Modular Actor Formalism for Artificial Intelligence IJCAI 1973; G. Agha. ACTORS: A Model of Concurrent Computation in Distributed Systems. The MIT Press: Cambridge, MA, 1986*) were proposed as universal primitives of concurrent computation.
Actors are self-contained, interactive autonomous components of a computing system that communicate by asynchronous message passing.
The basic actor primitives are:

- *create*: creating an actor from a behavior description and a set of parameters, possibly including existing actors;
- *send*: sending a message to an actor;
- *become*: changing an actor's local state.

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